AGREEGATE FUNCTIONS:

Program:

```sql
-- Step 1: Create Database and Table
CREATE DATABASE aggregate_bank_db;
USE aggregate_bank_db;
CREATE TABLE bank_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
-- Step 2: Insert Data
INSERT INTO bank_transactions VALUES
(101,'Tejaswini','Hyderabad','Deposit',5000,'2024-01-05'),
(102,'Chinni','Hyderabad','Withdrawal',2000,'2024-01-06'),
(103,'Sree','Vijayawada','Deposit',12000,'2024-01-08'),
(104,'Latha','Vizag','Deposit',8000,'2024-01-10'),
(105,'Ravi','Hyderabad','Withdrawal',3500,'2024-01-11'),
(106,'Praneeth','Vizag','Deposit',15000,'2024-01-12'),
(107,'Sahasra','Vijayawada','Withdrawal',1000,'2024-01-13'),
(108,'Bhavani','Hyderabad','Deposit',9000,'2024-01-14'),
(109,'Nikila','Vizag','Withdrawal',4000,'2024-01-15'),
(110,'Karthik','Vijayawada','Deposit',11000,'2024-01-16');
```

Output:

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:03:45	CREATE DATABASE aggregate_bank_db	1 row(s) affected	0.015 sec
2	15:03:47	USE aggregate_bank_db	0 row(s) affected	0.000 sec
3	15:03:51	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, ...	0 row(s) affected	0.031 sec
4	15:04:02	INSERT INTO bank_transactions VALUES (101,'Tejaswini','Hyderaba...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.015 sec

Program:

```sql
-- Step 3: SUM()
SELECT SUM(amount) AS Total_Amount
FROM bank_transactions;
```

Output:

Result Grid

Total_Amount
70500.00

5 15:06:07 SELECT SUM(amount) AS Total_Amount FROM bank_transactions ... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```sql
-- Step 4: AVG()
SELECT AVG(amount) AS Average_Transaction
FROM bank_transactions;
```

Output:

Result Grid Filter

Average_Transaction
7050.000000

6 15:06:49 SELECT AVG(amount) AS Average_Transaction FROM bank_transa... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```sql
-- Step 5: MAX()
SELECT MAX(amount) AS Highest_Transaction
FROM bank_transactions;
```

Output:

Result Grid Filter R

Highest_Transaction
15000.00

7 15:07:33 SELECT MAX(amount) AS Highest_Transaction FROM bank_trans... 1 row(s) returned 0.016 sec / 0.000 sec

Program:

```sql
-- Step 6: MIN()
SELECT MIN(amount) AS Lowest_Transaction
FROM bank_transactions;
```

Output:

Result Grid | Filter Rows:

Lowest_Transaction
1000.00

8 15:08:20 SELECT MIN(amount) AS Lowest_Transaction FROM bank_trans... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```sql
-- Step 7: COUNT()
SELECT COUNT(*) AS Total_Transactions
FROM bank_transactions;
```

Output:

Result Grid | Filter R

Total_Transactions
10

9 15:09:12 SELECT COUNT(*) AS Total_Transactions FROM bank_transactio... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```sql
-- Step 8: WHERE + SUM()
SELECT SUM(amount) AS Total_Deposit
FROM bank_transactions
WHERE transaction_type = 'Deposit';
```

Output:

Result Grid

Total_Deposit
60000.00

10 15:09:55 SELECT SUM(amount) AS Total_Deposit FROM bank_transaction... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```sql
-- Step 9: GROUP BY + SUM()
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output:

Result Grid | Filter Rows:

	branch_name	Total_Amount
▶	Hyderabad	19500.00
	Vijayawada	24000.00
	Vizag	27000.00

11	15:10:36	SELECT branch_name, SUM(amount) AS Total_Amount FRO...	3 row(s) returned	0.000 sec / 0.000 sec

Program:

```sql
-- Step 10: GROUP BY + HAVING
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name
HAVING SUM(amount) > 20000;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	branch_name	Total_Amount
▶	Vijayawada	24000.00
	Vizag	27000.00

Result Grid | Form Editor | Field Types | Query Stats

Result 8 × | Read Only | Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	15:07:33	SELECT MAX(amount) AS Highest_Transaction FROM bank_trans...	1 row(s) returned	0.016 sec / 0.000 sec
8	15:08:20	SELECT MIN(amount) AS Lowest_Transaction FROM bank_trans...	1 row(s) returned	0.000 sec / 0.000 sec
9	15:09:12	SELECT COUNT(*) AS Total_Transactions FROM bank_transactio...	1 row(s) returned	0.000 sec / 0.000 sec
10	15:09:55	SELECT SUM(amount) AS Total_Deposit FROM bank_transaction...	1 row(s) returned	0.000 sec / 0.000 sec
11	15:10:36	SELECT branch_name, SUM(amount) AS Total_Amount FRO...	3 row(s) returned	0.000 sec / 0.000 sec
12	15:12:12	SELECT branch_name, SUM(amount) AS Total_Amount FRO...	2 row(s) returned	0.000 sec / 0.000 sec

Program:

```sql
-- Step 11: GROUP BY + ORDER BY
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name
ORDER BY Total_Amount DESC;
```

Output:

Result Grid | Filter Rows:

branch_name	Total_Amount
Vizag	27000.00
Vijayawada	24000.00
Hyderabad	19500.00

13 15:12:51 SELECT branch_name, SUM(amount) AS Total_Amount FRO... 3 row(s) returned 0.000 sec / 0.000 sec

Program:

```sql
-- Step 12: GROUP BY + HAVING + ORDER BY
SELECT branch_name,
       COUNT(*) AS Total_Transactions
FROM bank_transactions
GROUP BY branch_name
HAVING COUNT(*) >= 3
ORDER BY Total_Transactions DESC;
```

Output:

Result Grid | Filter Rows:

branch_name	Total_Transactions
Hyderabad	4
Vijayawada	3
Vizag	3

14 15:14:24 SELECT branch_name, COUNT(*) AS Total_Transactions FR... 3 row(s) returned 0.000 sec / 0.000 sec

Program:

```sql
-- Step 13: WHERE + COUNT()
SELECT COUNT(*) AS Withdrawals
FROM bank_transactions
WHERE transaction_type = 'Withdrawal';
```

Output:

Result Grid

Withdrawals
4

15 15:15:19 SELECT COUNT(*) AS Withdrawals FROM bank_transactions WH... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```sql
-- Step 14: WHERE + AVG()
SELECT AVG(amount) AS Avg_Deposit
FROM bank_transactions
WHERE transaction_type = 'Deposit';
```

Output:

Result Grid

Avg_Deposit
10000.000000

16 15:16:06 SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions ... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```sql
-- Step 15: GROUP BY + MAX()
SELECT branch_name,
       MAX(amount) AS Highest_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output:

Result Grid Filter Rows:

branch_name	Highest_Amount
Hyderabad	9000.00
Vijayawada	12000.00
Vizag	15000.00

17 15:17:51 SELECT branch_name, MAX(amount) AS Highest_Amount FR... 3 row(s) returned 0.000 sec / 0.000 sec

Program:

```sql
-- Step 16: GROUP BY + MIN()
SELECT branch_name,
       MIN(amount) AS Lowest_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output:

Result Grid | Filter Rows:

branch_name	Lowest_Amount
Hyderabad	2000.00
Vijayawada	1000.00
Vizag	4000.00

18 15:18:48 SELECT branch_name, MIN(amount) AS Lowest_Amount FR... 3 row(s) returned 0.000 sec / 0.000 sec

Program:

```sql
-- Step 17: WHERE + ORDER BY
SELECT *
FROM bank_transactions
WHERE amount > 8000
ORDER BY amount DESC;
```

Output:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	Praneeth	Vizag	Deposit	15000.00	2024-01-12
103	Sree	Vijayawada	Deposit	12000.00	2024-01-08
110	Karthik	Vijayawada	Deposit	11000.00	2024-01-16
108	Bhavani	Hyderabad	Deposit	9000.00	2024-01-14
NULL	NULL	NULL	NULL	NULL	NULL

#	Time	Action	Message	Duration / Fetch
14	15:14:24	SELECT branch_name, COUNT(*) AS Total_Transactions FR...	3 row(s) returned	0.000 sec / 0.000 sec
15	15:15:19	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WH...	1 row(s) returned	0.000 sec / 0.000 sec
16	15:16:06	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions ...	1 row(s) returned	0.000 sec / 0.000 sec
17	15:17:51	SELECT branch_name, MAX(amount) AS Highest_Amount FR...	3 row(s) returned	0.000 sec / 0.000 sec
18	15:18:48	SELECT branch_name, MIN(amount) AS Lowest_Amount FR...	3 row(s) returned	0.000 sec / 0.000 sec
19	15:19:39	SELECT * FROM bank_transactions WHERE amount > 8000 OR...	4 row(s) returned	0.000 sec / 0.000 sec

Program:

```
-- Step 18: GROUP BY + COUNT()
SELECT branch_name,
        COUNT(*) AS Customer_Count
FROM bank_transactions
GROUP BY branch_name;
```

Output:

branch_name	Customer_Count
Hyderabad	4
Vijayawada	3
Vizag	3

#	Time	Action	Message	Duration / Fetch
20	15:20:28	SELECT branch_name, COUNT(*) AS Customer_Count FROM...	3 row(s) returned	0.000 sec / 0.000 sec

Program:

```sql
-- Step 19: HAVING + COUNT()
SELECT branch_name,
        COUNT(*) AS Total_Transactions
FROM bank_transactions
GROUP BY branch_name
HAVING COUNT(*) > 3;
```

Output:

Result Grid | Filter Rows:

branch_name	Total_Transactions
Hyderabad	4

21 15:21:16 SELECT branch_name, COUNT(*) AS Total_Transactions FR... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```sql
-- Step 20: GROUP BY + AVG()
SELECT branch_name,
        AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output:

Result Grid | Filter Rows:

branch_name	Avg_Amount
Hyderabad	4875.000000
Vijayawada	8000.000000
Vizag	9000.000000

22 15:21:59 SELECT branch_name, AVG(amount) AS Avg_Amount FROM... 3 row(s) returned 0.015 sec / 0.000 sec

Program:

```sql
-- Step 21: WHERE + GROUP BY + HAVING
SELECT branch_name,
        SUM(amount) AS Total_Deposit
FROM bank_transactions
WHERE transaction_type = 'Deposit'
GROUP BY branch_name
HAVING SUM(amount) > 15000;
```

Output:

Result Grid | Filter Rows:

branch_name	Total_Deposit
Vijayawada	23000.00
Vizag	23000.00

23 15:22:47 SELECT branch_name, SUM(amount) AS Total_Deposit FRO... 2 row(s) returned 0.000 sec / 0.000 sec

Program:

```sql
-- Step 22: GROUP BY + HAVING + ORDER BY
SELECT branch_name,
       AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name
HAVING AVG(amount) > 7000
ORDER BY Avg_Amount DESC;
```

Output:

Result Grid | Filter Rows:

branch_name	Avg_Amount
Vizag	9000.000000
Vijayawada	8000.000000

24 15:23:29 SELECT branch_name, AVG(amount) AS Avg_Amount FROM... 2 row(s) returned 0.000 sec / 0.000 sec